INCH-POUND

MIL-DTL-2087E
01 December 2023

SUPERSEDING
MIL-DTL-2087D (AR)
26 June 2019

DETAIL SPECIFICATION

DEMOLITION KIT, BANGALORE TORPEDO: M1A2
and
DEMOLITION KIT, BANGALORE TORPEDO: 2.5 FEET M1A3
LOADING, ASSEMBLING AND PACKING

Reactivated after 30 November 2023, may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling, and packing for the following items:

DEMOLITION KIT, BANGALORE TORPEDO: M1A2
DEMOLITION KIT, BANGALORE TORPEDO: 2.5 FT, M1A3

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	–	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	–	DOD Preferred Method for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT CENTER DRAWINGS

8863369	–	Demolition Kit, Bangalore Torpedo, M1A2
8863370	–	Loading Assembly, Bangalore Torpedo, M1A2
8863371	–	Sleeve, Nose
8863372	–	Sleeve, Connecting
8863374	–	Tube, Bangalore Torpedo, M1A2
8863375	–	End Cap and Detonator Well Assembly
8863376	–	End Cap and Detonator Well Assembly, Alternate
8863378	–	Cap, End
8863379	–	Well, Detonator
8863380	–	Cap, End, Alternate
8863381	–	Well, Detonator, Alternate
9216100	–	Packing and Marking for Demolition Kit, Bangalore Torpedo, M1A2
9266746	–	Booster
13008483	–	Packing and Marking for Demolition Kit, Bangalore Torpedo, 2.5 FT M1A3
13010551	–	Combination of Adopted Items for: Demolition Kit, Bangalore Torpedo: 2.5 FT, M1A3
13010552	–	Demolition Kit, Bangalore Torpedo, 2.5 FT, M1A3
13010553	–	Loading Assembly, Bangalore Torpedo, M1A3
13101293	–	Tube, Bangalore Torpedo, M1A3
13113421	–	Combination of Adopted Items for: Demolition Kit, Bangalore Torpedo, M1A2

(Copies of the above drawings are available by email at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil.)

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US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

19-48-4116/125B	–	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets
19-48-4138/117	–	Unitization Procedures for Boxed Ammunition and Components on Skidded Bases

(Copies of these documents may be requested online at https://www.dau.edu/cop/ammo/Pages/Default.aspx.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1742/E1742M	–	Standard Practice for Radiographic Examination

(Copies of the above document can be obtained online at https://www.astm.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article inspection. When specified in (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Materials, components, and assemblies. The M1A2 and M1A3 Bangalore Torpedo Demolition Kits shall conform to Drawings 8863369, 13010552, and their subsidiary drawings.

3.2.1 Specific gravity of booster. The specific gravity of the booster shall be not less than 1.55 grams per cubic centimeter (gm/cc).

3.2.2 Waterproofness of tube. The tube shall withstand an internal air pressure of not less than five (5) pounds per square inch gage (psig) without any evidence of leakage.

3.2.3 Waterproofness of end cap and detonator well assembly. The end cap and detonator

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well assembly shall withstand an internal air pressure of not less than fifteen (15) pounds per square inch gage (psig) without any evidence of leakage.

3.3 Functioning. When initiated by a standard military electric or non-electric blasting cap (see 6.2) the Bangalore Torpedo loading assembly shall detonate high order and propagate through a connecting sleeve to another Bangalore Torpedo loading assembly (see 6.3).

3.4 Explosive fill. Cavitation or voids in the loading assembly shall be not greater than ¼ inch depth and 1 inch width in any direction. Cavitation shall be permitted only on the pouring side.

3.5 Dents. The tube shall show no dents within four (4) inches of a flange. When greater than four (4) inches from a flange the tube shall show not greater than four (4) dents, or any one (1) dent greater than 1/8 inch.

3.6 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.7 Workmanship. All parts, sub-assemblies, and assemblies shall be clean and free of burrs, chips, sharp edges, cracks, unblended radii, burn marks, checks, sinks, crazes, blisters, excess flash, dirt, grease, oil, rust, corrosion, and other foreign matter.

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4. VERIFICATION

TABLE I. Requirements/verification cross reference matrix.

Method of Verification 1 - Analysis 3 - Examination 2 - Demonstration 4 - Test		Classes of Verification A - First article B - Conformance						
Section 3 and 5 Requirements	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article inspection	X		X	X	X		4.2
3.1.2	Conformance inspection	X		X	X		X	4.3
3.2	Materials, components, and assemblies			X	X	X	X	4.3.4.1 through 4.3.4.14
3.2.1	Specific gravity of booster	X			X	X	X	4.4.1
3.2.2	Waterproofness of tube			X	X	X	X	4.4.4
3.2.3	Waterproofness of end cap and detonator well assembly			X	X	X	X	4.4.5
3.3	Functioning				X	X	X	4.4.2
3.4	Explosive fill			X		X		4.4.3
3.5	Dents			X		X	X	4.3.4.9
3.6	Ammunition lot numbering			X		X	X	4.3.3, 4.3.4.14, 4.3.4.16
3.7	Workmanship			X		X	X	4.3.4.1 through 4.3.4.17
5.1	Packaging and marking			X		X	X	4.3.4.15 through 4.3.4.16
5.1.1	Unitization			X			X	4.3.4.17

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2)
b. Conformance Inspection (see 4.3)

4.2 First article inspection. When specified, a sample shall be subjected to first article inspections in accordance with Table II. First article samples shall be manufactured using the same vendors, methods, materials, equipment, processes, procedures, and facilities as will be used during regular production.

4.2.1 First article rejection. If one (1) or more assemblies, subassemblies, materials, or components fail to comply with the first article requirements, the sample(s) and first article shall be rejected.

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TABLE II. First article inspections and tests.

Examination or test	Drawing(s)	Sample Size	Requirement Paragraph	Paragraph Reference / Inspection Method
End cap / alt.	8863378/8863380	15	3.2	4.3.4.1/4.3.4.2
Detonator well / alt.	8863379/8863381	15	3.2	4.3.4.3/4.3.4.4
End cap and detonator well assembly / alt.	8863375/8863376	15	3.2	4.3.4.5/4.3.4.6
Nose sleeve	8863371	15	3.2	4.3.4.7
Connecting sleeve	8863372	15	3.2	4.3.4.8
Tube	8863374 (5 ft.) 13101293 (2.5 ft.)	15 1/	3.2	4.3.4.9
Booster Examination for defects Specific gravity	9266746	 15 15	 3.2 3.2.1	 4.3.4.10 4.4.1
Loading assembly Examination for defects Explosive fill	8863370 (5 ft.) 13010553 (2.5 ft)	1/ 40 40	 3.2 3.4	 4.3.4.11 through 4.3.4.14 4.4.3
Functioning	8863369 (5 ft.) 13010552 (2.5 ft)	20 test sets 1/	3.3	4.4.2
Packaging and marking	9216100 (5 ft.) 13008483(2.5 ft)	1 box	5.1	4.3.4.15, 4.3.4.16
Note: 1/ Quantity is for either the 5 ft or 2.5 ft.				

4.3 Conformance inspection.

4.3.1 Conformance rejection. If any assembly, subassembly, material, or component fails to comply with the conformance inspection requirements of 4.3 and its subparagraphs, the lot from which the sample was taken shall be rejected.

4.3.2 Lot formation. Inspection lots shall conform to the requirements of MIL-STD-1916 (when applicable). When not applicable, lot formation shall be homogeneous. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings and revisions, and the same specifications and revisions. In addition, each lot shall contain materials from not more than one lot interfix number from one manufacturer except the following:

a. Composition A-3 from not more than two lots from one lot interfix number from one manufacturer.
b. Composition B-4 from not more than two lots from one lot interfix number from one manufacturer which contains TNT from not more than two lots from one lot

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interfix number from one manufacturer.

4.3.3 Ammunition lot number. Visually verify that a unique ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

4.3.4 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraphs 4.3.4.1 through 4.3.4.17, the definitions of critical, major, and minor defects are provided in paragraph "Definitions" of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph "Acceptance by contractor-proposed provisions" of MIL-STD-1916).

4.3.4.1 Cap, end.			Drawing Number 8863378 Next Higher Assembly 8863375	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	True position of center hole	IV	3.2	Gage
102	Pitch diameter of thread	IV	3.2	Gage
103	Minor diameter of thread	IV	3.2	Gage
104	Small inside diameter below thread	IV	3.2	Gage
105	Large outside diameter	IV	3.2	Gage
106	Second largest diameter	IV	3.2	Gage
107	Length of cavity	IV	3.2	Gage
108	Crack or split	IV	3.2	Visual
Minor				
201	Workmanship	II	3.7	Visual

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4.3.4.2 Cap, end, alternate.			Drawing Number 8863380 Next Higher Assembly 8863376	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	True position of center hole	IV	3.2	Gage
102	Large outside diameter	IV	3.2	Gage
103	Second largest diameter	IV	3.2	Gage
104	Crack or split	IV	3.2	Visual
Minor				
201	Workmanship	II	3.7	Visual

4.3.4.3 Well, detonator.			Drawing Number 8863379 Next Higher Assembly 8863375	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Inside diameter of well	IV	3.2	Gage
102	Outside diameter of well	IV	3.2	Gage
103	Thickness of flange	IV	3.2	Gage
104	Diameter of flange	IV	3.2	Gage
105	True position of flange diameter	IV	3.2	Gage
106	Crack or split	IV	3.2	Visual
Minor				
201	Workmanship	II	3.7	Visual

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4.3.4.4 Well, detonator, alternate.			Drawing Number 8863381 Next Higher Assembly 8863376	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Pitch diameter of thread	IV	3.2	Gage
102	Minor diameter of thread	IV	3.2	Gage
103	Large outside diameter	IV	3.2	Gage
104	Inside diameter of well	IV	3.2	Gage
105	Outside diameter of well	IV	3.2	Gage
106	Crack or split	IV	3.2	Visual
Minor				
201	Workmanship	II	3.7	Visual

4.3.4.5 End cap and detonator well assembly.			Drawing Number 8863375 Next Higher Assembly 8863370 (5 ft.) 13010553 (2.5 ft.)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	True position of inside diameter of detonator well	IV	3.2	Gage
102	Foreign matter on interior surfaces of threads or detonator well	IV	3.2	Visual
103	Protective finish for energetic surfaces omitted or incomplete	IV	3.2	Visual
104	Waterproofness	IV	3.2.3	4.4.5
Minor				
201	Workmanship	II	3.7	Visual

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4.3.4.6 End cap and detonator well assembly, alternate.			Drawing Number 8863376 Next Higher Assembly 8863370 (5 ft.) 13010553 (2.5 ft.)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	True position of inside diameter of detonator well	IV	3.2	Gage
102	Detonator well not flush or above reference surface of end cap	IV	3.2	Gage
103	Juncture of end cap and detonator well not clinched full 360 degrees or tight around entire periphery	IV	3.2	Manual / Visual
104	Soldering of juncture of end cap and detonator well omitted or incomplete	IV	3.2	Visual
105	Foreign matter on interior surfaces of threads or detonator well	IV	3.2	Visual
106	Protective finish for energetic surfaces omitted or incomplete	IV	3.2	Visual
107	Waterproofness	IV	3.2.3	4.4.5
Minor				
201	Workmanship	II	3.7	Visual

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4.3.4.7 Sleeve, nose.			Drawing Number 8863371 Next Higher Assembly 8863369 (5 ft.) 13010552 (2.5 ft.)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Large inside diameter	IV	3.2	Gage
102	Wall thickness (4 places)	IV	3.2	Gage
103	Locking tab omitted, improperly positioned, or not bent	IV	3.2	Visual
104	Location of locking tab opening from open end of nose	IV	3.2	Gage
105	Length of locking tab opening	IV	3.2	Gage
106	Protective finish omitted or incomplete	IV	3.2	Visual
Minor				
201	Workmanship	II	3.7	Visual

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4.3.4.8 Sleeve, connecting.			Drawing Number 8863372 Next Higher Assembly 8863369 (5 ft.) 13010552 (2.5 ft.)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Large inside diameter	IV	3.2	Gage
102	Wall thickness (4 places)	IV	3.2	Gage
103	Overall length	IV	3.2	Gage
104	Locking tabs omitted or improperly positioned (6 places)	IV	3.2	Visual
105	Locking tab depth from outside surface (6 places)	IV	3.2	Gage
106	Stopping tab not bent or improperly positioned (x3)	IV	3.2	Visual
107	Width of stopping tab (x3)	IV	3.2	Gage
108	Protective finish omitted or incomplete	IV	3.2	Visual
Minor				
201	Workmanship	II	3.7	Visual

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4.3.4.9 Tube, Bangalore Torpedo, M1A2. Tube, Bangalore Torpedo, M1A3.			Drawing Number 8863374 13101293 Next Higher Assembly 8863374 (5 ft.) 13101293 (2.5 ft.)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	True position of flange (x2)	IV	3.2	Gage
102	Thickness of flange (x2)	IV	3.2	Gage
103	Wall thickness (4 places)	IV	3.2	Gage
104	Large outside diameter	IV	3.2	Gage
105	Diameter of flange (x2)	IV	3.2	Gage
106	Total length	IV	3.2	Gage
107	Inside diameter (x2)	IV	3.2	Gage
108	Evidence of welding burrs, when applicable	IV	3.2	Visual
109	Dents	IV	3.5	Visual / Gage
110	Protective finish omitted or incomplete	IV	3.2	Visual
111	Waterproofness (alternative welded or buttwelded material only)	IV	3.2.2	4.4.4
Minor				
201	Workmanship	II	3.7	Visual

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4.3.4.10 Booster.			Drawing Number 9266746 Next Higher Assembly 8863370 (5 ft.) 13010553 (2.5 ft.)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	True position of detonator well cavity	IV	3.2	Gage
102	Diameter of detonator well cavity	IV	3.2	Gage
103	Depth of end cap and detonator well cavity	IV	3.2	Gage
104	Outside diameter	IV	3.2	Gage
105	Overall length	IV	3.2	Gage
106	Inside diameter of end cap counterbore	IV	3.2	Gage
107	Specific gravity	1/	3.2.1	4.4.1
Minor				
201	Workmanship	II	3.7	Visual
Notes: 1/ Three (3) samples per production shift per tooling.				

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4.3.4.11 Assembly, loading (prior to pouring and assembly).			Drawing Number 8863370 (5 ft.) 13010553 (2.5 ft.) Next Higher Assembly 9216100 (5 ft.) 13008483 (2.5 ft)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Waterproofness compound missing or inadequate (non-pouring end)	IV	3.2	Visual
102	Foreign matter on interior surfaces of threads or detonator well	IV	3.2	Visual
103	Filler or booster missing (non-pouring end)	IV	3.2	Visual
Minor				
201	Workmanship	II	3.7	Visual

4.3.4.12 Assembly, loading (prior to insertion of booster at pouring end).			Drawing Number 8863370 (5 ft.) 13010553 (2.5 ft.) Next Higher Assembly 9216100 (5 ft.) 13008483 (2.5 ft)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Depth to end of main charge	IV	3.2	Gage
102	Depth of cavity at booster interface exceeds ¼ inch	IV	3.2	Gage
103	Width of cavity at booster interface exceeds one inch in any direction	IV	3.2	Gage
Minor				
201	Workmanship	II	3.7	Visual

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4.3.4.13 Assembly, loading (prior to assembly of end cap and detonator well assemblies).			Drawing Number 8863370 (5 ft.) 13010553 (2.5 ft.) Next Higher Assembly 9216100 (5 ft.) 13008483 (2.5 ft)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Booster omitted or improper	IV	3.2	Visual
102	Felt washer omitted (if applicable)	IV	3.2	Visual
Minor				
201	Workmanship	II	3.7	Visual

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4.3.4.14 Assembly, loading.			Drawing Number 8863370 (5 ft.) 13010553 (2.5 ft.) Next Higher Assembly 9216100 (5 ft.) 13008483 (2.5 ft)	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Total length	IV	3.2	Gage
102	Outside diameter (both ends)	IV	3.2	Gage
103	360 degrees curling or crimping omitted or improper (both ends)	IV	3.2	Visual / Manual
104	End cap seal omitted or improper (both ends)	IV	3.2	Visual
105	Protective finish omitted or incomplete on ends	IV	3.2	Visual
106	Foreign matter on interior surfaces of threads or detonator well	IV	3.2	Visual
107	Threads damaged (either end)	IV	3.2	Visual
108	Functioning	15 test sets 1/	3.3	4.4.2
Minor				
201	Marking missing, incorrect, or illegible	II	3.2, 3.6	Visual
202	Workmanship	II	3.7	Visual
Note: 1/ Quantity is for either the 5 ft or 2.5 ft test sets.				

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4.3.4.15 Box, packing (prior to sealing).			Drawing Number 9216100 (5 ft.) 13008483 (2.5 ft.) Next Higher Assembly 19-48-4116/125B 19-48-4138/117	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Component or assembly missing or damaged	II	5.1	Visual
102	Component or assembly incorrectly packed	II	5.1	Visual
103	Pull straps not properly threaded	II	5.1	Visual
104	Wood preservative and heat treatment markings missing, incorrect, illegible, or not in proper location	II	5.1	Visual
Minor				
201	Workmanship	I	3.7	Visual

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4.3.4.16 Box, packing (sealed).			Drawing Number 9216100 (5 ft.) 13008483 (2.5 ft.) Next Higher Assembly 19-48-4116/125B 19-48-4138/117	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Box broken or split	II	5.1	Visual
102	Contents exposed	II	5.1	Visual
103	Strapping missing, damaged, or loose	II	5.1	Visual / Manual
104	Hardware missing, damaged, or loose	II	5.1	Visual / Manual
105	Contents loose when box is tilted	II	5.1	Manual
Minor				
201	Marking missing, incorrect, or illegible	I	3.6, 5.1	Visual
202	Strapping improperly assembled	I	5.1	Visual
203	Hardware improperly engaged	I	5.1	Visual / Manual
204	Metallic seal missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly	I	5.1	Visual / Manual
205	Workmanship	I	3.7	Visual

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4.3.4.17 Unitization.			Drawing Number 19-48-4116/125B 19-48-4138/117 Next Higher Assembly	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location	II	5.1.1	Visual
102	Pallet load markings -including 2D bar codes- missing, incorrect, illegible, or not in proper location	II	5.1.1	Visual
103	Filler assembly not built, treated, or marked correctly	II	5.1.1	Visual
104	Boxes incorrectly oriented or positioned on pallet	II	5.1.1	Visual
105	Strapping not applied correctly or incorrect style	II	5.1.1	Visual
106	Unit load not tightly strapped or any strap loose	II	5.1.1	Visual
Minor				
201	Workmanship	I	3.7	Visual

4.4 Test methods and procedures.

4.4.1 Specific gravity of booster. If any booster fails to meet the specific gravity, the samples shall be rejected since the last successful specific gravity sample. The booster shall be weighed on a balance accurate to 0.1 grams and the weight recorded as dry weight. The booster shall then be weighed while completely submerged in distilled water, taking care to remove all air bubbled remaining on the surfaces of the booster. The temperature of the water shall be 68 ºF ± 2 ºF. This weight shall be recorded as weight immersed in water. When submerged, the position of the booster shall be such that no air is entrapped. All booster samples shall be thoroughly dried prior to being added to assembly. Specific gravity shall be calculated as follows:

$$Specific\ gravity = \frac{Weight\ of\ booster\ dry}{\left(\begin{matrix}Weight\ of\\ booster\\ dry\end{matrix}\right) - \left(\begin{matrix}Weight\ of\ booster\\ immersed\ in\ water\end{matrix}\right)} \times \begin{matrix}Specific\ gravity\ of\ H_2O\\ at\ noted\ temperature\\ (See\ TABLE\ III)\end{matrix}$$

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TABLE III. Specific gravity of water within temperature range.

Temp °F	Specific Gravity of H_2O
66	0.9984
67	0.9983
68	0.9982
69	0.9981
70	0.9980

4.4.2 Functioning. A test set shall consist of two loaded assemblies attached by a connecting sleeve and a nose at one end (reference Drawing 8863369 and 13010552 for the assembled M1A2 and M1A3, respectively). The test set shall be detonated by initiation either by a standard military electric or non-electric blasting cap (see 6.2) held in place with the priming adapter. Any test set which fails to detonate high order (see 6.3) shall reject the sample(s) and lot.

4.4.3 Explosive fill. Radiographic equipment, operations and procedures shall be in accordance with ASTM E1742/E1742M. The radiographic examination shall detect voids, cracks, cavitation, separation, and cracking of the explosive and presence of any foreign material. The assembly shall be radiographically inspected two (2) times. The assembly shall be inspected along its length, rotated approximately ninety degrees (90°) on its cylindrical axis, and inspected a second time. Any assembly which does not meet the criteria shall reject the sample(s) and first article testing.

4.4.4 Waterproofness of tube. The tube shall be assembled to the applicable test equipment and the specified internal air pressure applied for not less than five (5) seconds while the entire assembly is submerged under not less than two (2) inches of water. Any assembly which shows evidence of leakage by emitting air bubbles at any point other than where the test equipment assembles to the tube shall reject the sample(s) and lot.

4.4.5 Waterproofness of end cap and detonator well assembly. The assembly shall be assembled to the applicable test equipment and the specified internal air pressure applied for not less than five (5) seconds, while the entire assembly is submerged under not less than two (2) inches of water. Any assembly which shows evidence of leakage by emitting air bubbles at any point other than where the test equipment assembles to the assembly, shall reject the sample(s) and lot.

5. PACKAGING

5.1 Packaging and marking. The materials, components, and assemblies shall comply with Drawing 13113421 for the M1A2 and Drawing 13010551 for the M1A3 and all their subsidiary drawings.

5.1.1 Unitization. Palletize M1A2 in accordance with 19-48-4138/117 and M1A3 in accordance with 19-48-4116/125B and their subsidiary drawings.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Bangalore Torpedo Demolition Kit is primarily employed for clearing paths through wire obstacles and heavy undergrowth.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and of all reference documents cited in Section 2 and applicable documents from Section 6.
b. Requirement/CDRL for Physical Configuration Audit (PCA).
c. Requirement/CDRL for First article inspection.
d. Requirement/CDRL for Conformance inspection.
e. Requirement/CDRL for Ammunition Data Cards (ADC) in accordance with MIL-STD-1168.
f. Requirement/CDRL for Propellants and Explosives data in accordance with MIL-STD-1171.
g. Requirement/CDRL for Measurement System Evaluation (MSE) and associated test procedures. Military blasting caps (equivalent to Drawing 13026127) used during testing should be submitted as part of the AIE.
h. Bangalore Torpedo Demolition Kit type.

6.3 High order detonation. The complete detonation of the Bangalore Torpedo is evidenced by the propagation of the blasting cap and from the blasting cap to the loading assembly. The loading assembly detonates high order, fully consuming all explosive material.

6.4 Overview of changes. The following approved Engineering Change Proposal (ECP) is being incorporated into this document: R23Q2028.

6.5 Subject term (key word) listing.
Demolitions

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Army – AR
Navy - MC

Preparing Activity:
Army – AR
(Project 1375-2024-007)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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